

Cloud Computing Lab — Experiment 2

Connecting Two Virtual Machines Using a VirtualBox NAT Network

Field	Details
Name	TODO: Add your name
Roll Number	TODO: Add your roll number
Section	BTECH_CSE_SECB_2023
Course	Cloud Computing Lab
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1. Aim

Create two virtual machines, create a custom **NAT Network** in Oracle VirtualBox, attach both VMs to it, and verify that the VMs can communicate with each other using `ping`.

2. Theory

VirtualBox offers several networking modes for a VM's virtual network adapter:

Mode	VM ↔ Internet	VM ↔ VM	Host ↔ VM
NAT (default)	Yes	No — every VM sits behind its own private NAT router	Port forwarding only
NAT Network	Yes	Yes — all VMs on the network share one virtual router and subnet	Port forwarding only
Bridged Adapter	Yes	Yes	Yes
Host-only Adapter	No	Yes	Yes
Internal Network	No	Yes	No

With plain **NAT**, VirtualBox gives each VM its own isolated virtual router, so every VM receives the same address (`10.0.2.15`) and the VMs cannot reach one another. A **NAT Network** is a shared virtual switch plus router: VirtualBox's DHCP server hands each VM a unique address from one subnet (here `192.168.50.0/24`), so the VMs can talk to each other while still reaching the Internet through NAT. In a cloud this corresponds to placing instances in the same private subnet of a **VPC** (Virtual Private Cloud).

On a VirtualBox NAT Network the first addresses are reserved: `.1` is the gateway, `.2` the DHCP server, and `.3` the NAT engine's built-in DNS proxy.

3. Requirements

Item	Value
Host OS	Windows 11 Home Single Language, 64-bit (Intel Core i7-12700H, 16 GB RAM)
Hypervisor	Oracle VirtualBox 7.2.18
Guest OS (both VMs)	Ubuntu 26.04.1 LTS (kernel 7.0.0-31-generic)

4. Configuration

Virtual machines

Setting	VM 1	VM 2
Name	Ubuntu-Cloud-Lab	Ubuntu-Cloud-Lab-2 (full clone of VM 1)
Hostname	ubuntu-cloud-lab	ubuntu-cloud-lab-2
CPU / RAM	2 vCPU / 4096 MB	2 vCPU / 4096 MB
Adapter 1	Intel PRO/1000 MT Desktop, NAT Network CCloudLabNAT	Intel PRO/1000 MT Desktop, NAT Network CCloudLabNAT
MAC address	08:00:27:E0:A5:4F	08:00:27:27:6A:D1
IP on NAT (before)	10.0.2.15	10.0.2.15
IP on NAT Network (after)	192.168.50.5	192.168.50.4

NAT Network

Setting	Value
Name	CCloudLabNAT
IPv4 prefix	192.168.50.0/24
Gateway	192.168.50.1
DHCP	Enabled (server 192.168.50.2, pool 192.168.50.4 - .254)
IPv6	Disabled

The full configuration export is in [network-configuration.txt](#).

5. Procedure

1. **Create 2 VMs.** VM 1 (Ubuntu-Cloud-Lab) is the Ubuntu VM from Experiment 1. VM 2 was created as a full clone with **new MAC addresses** (*Machine* → *Clone*, MAC policy "Generate new MAC addresses"), and its hostname and machine-id were changed so the two VMs are distinct hosts on the network.

```
VBoxManage clonevm Ubuntu-Cloud-Lab --name Ubuntu-Cloud-Lab-2 --mode machine --register
```

```
# inside VM 2
sudo hostnamectl set-hostname ubuntu-cloud-lab-2
sudo rm /etc/machine-id && sudo systemd-machine-id-setup
```

2. **Check the IP addresses (ifconfig / ip a).** Started both VMs on the default NAT adapter and ran `ip a`. Both VMs showed the same address, `10.0.2.15/24`, because each one sits behind its own private NAT router.
3. **Create a custom NAT Network.** *File* → *Tools* → *Network* → *NAT Networks* → *Create*. Named it `CloudLabNAT`, IPv4 prefix `192.168.50.0/24`, **Enable DHCP** checked, then *Apply*.

```
VBoxManage natnetwork add --netname CloudLabNAT --network "192.168.50.0/24" --enable --dhcp on
```

4. **Attach VM 1 to the NAT Network.** *Machine* → *Settings* → *Expert* → *Network* → *Adapter 1* → *Attached to: "NAT Network"* → *Name: CloudLabNAT* → OK.
5. **Do the same for VM 2.**

```
VBoxManage modifyvm Ubuntu-Cloud-Lab --nic1 natnetwork --nat-network1 CloudLabNAT
VBoxManage modifyvm Ubuntu-Cloud-Lab-2 --nic1 natnetwork --nat-network1 CloudLabNAT
```

6. **Shut down both VMs and power them on again**, so that each one requests a new address from the NAT Network's DHCP server.
7. **Check the new IP addresses** with `ip a`: VM 1 → `192.168.50.5`, VM 2 → `192.168.50.4`.
8. **Ping the VMs from each other:**

```
# on VM 1
ping -c 5 192.168.50.4
# on VM 2
ping -c 5 192.168.50.5
```

Issue faced and fix

After the restart, VM 2 received `192.168.50.4`, but VM 1 stayed on "*getting IP configuration*". NetworkManager's log showed that the DHCP server offered `192.168.50.3`, and the duplicate-address check then reported a conflict for it:

```
dhcp4 (enp0s3): state changed new lease, address=192.168.50.3, acd conflict
```

`ip neigh` on VM 2 showed that `192.168.50.3` answers from the NAT engine's own MAC address (`52:54:00:12:35:00`), and the DHCP server configuration lists `192.168.50.3` as the network's DNS server (option 6). VirtualBox's DHCP pool for a new NAT Network starts at `.3`, which collides with its own DNS proxy address. The fix was to give VM 1 a DHCP reservation and move the pool so `.3` is never handed out:

```
VBoxManage dhcpserver modify --network=CloudLabNAT --vm=Ubuntu-Cloud-Lab --nic=1 --fixed-address=192.168.50.5
VBoxManage dhcpserver modify --network=CloudLabNAT --lower-ip=192.168.50.4 --upper-ip=192.168.50.254
VBoxManage dhcpserver restart --network=CloudLabNAT
```

After reconnecting the interface (`nmcli device disconnect enp0s3 && nmcli device connect enp0s3`), VM 1 received 192.168.50.5.

This also explains why the example in the assignment handout shows 100% packet loss. Its two VMs had the **same MAC address** (08:00:27:13:c3:94, a clone that kept its MAC), so they could not both work on the same network. Cloning with new MAC addresses avoids that problem.

6. Output

Before — both VMs on default NAT

```
student@ubuntu-cloud-lab:~$ ip a
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 ... state UP
    link/ether 08:00:27:e0:a5:4f brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3

student@ubuntu-cloud-lab-2:~$ ip a
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 ... state UP
    link/ether 08:00:27:27:6a:d1 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
```

After — both VMs on NAT Network `CloudLabNAT`

```
student@ubuntu-cloud-lab:~$ ip a show enp0s3; ping -c 5 192.168.50.4
    link/ether 08:00:27:e0:a5:4f brd ff:ff:ff:ff:ff:ff
    inet 192.168.50.5/24 brd 192.168.50.255 scope global dynamic noprefixroute enp0s3
PING 192.168.50.4 (192.168.50.4) 56(84) bytes of data.
64 bytes from 192.168.50.4: icmp_seq=1 ttl=64 time=0.748 ms
...
5 packets transmitted, 5 received, 0% packet loss, time 4143ms

student@ubuntu-cloud-lab-2:~$ ip a show enp0s3; ping -c 5 192.168.50.5
    link/ether 08:00:27:27:6a:d1 brd ff:ff:ff:ff:ff:ff
    inet 192.168.50.4/24 brd 192.168.50.255 scope global dynamic noprefixroute enp0s3
PING 192.168.50.5 (192.168.50.5) 56(84) bytes of data.
64 bytes from 192.168.50.5: icmp_seq=1 ttl=64 time=0.607 ms
...
5 packets transmitted, 5 received, 0% packet loss, time 4064ms
```

7. Screenshots

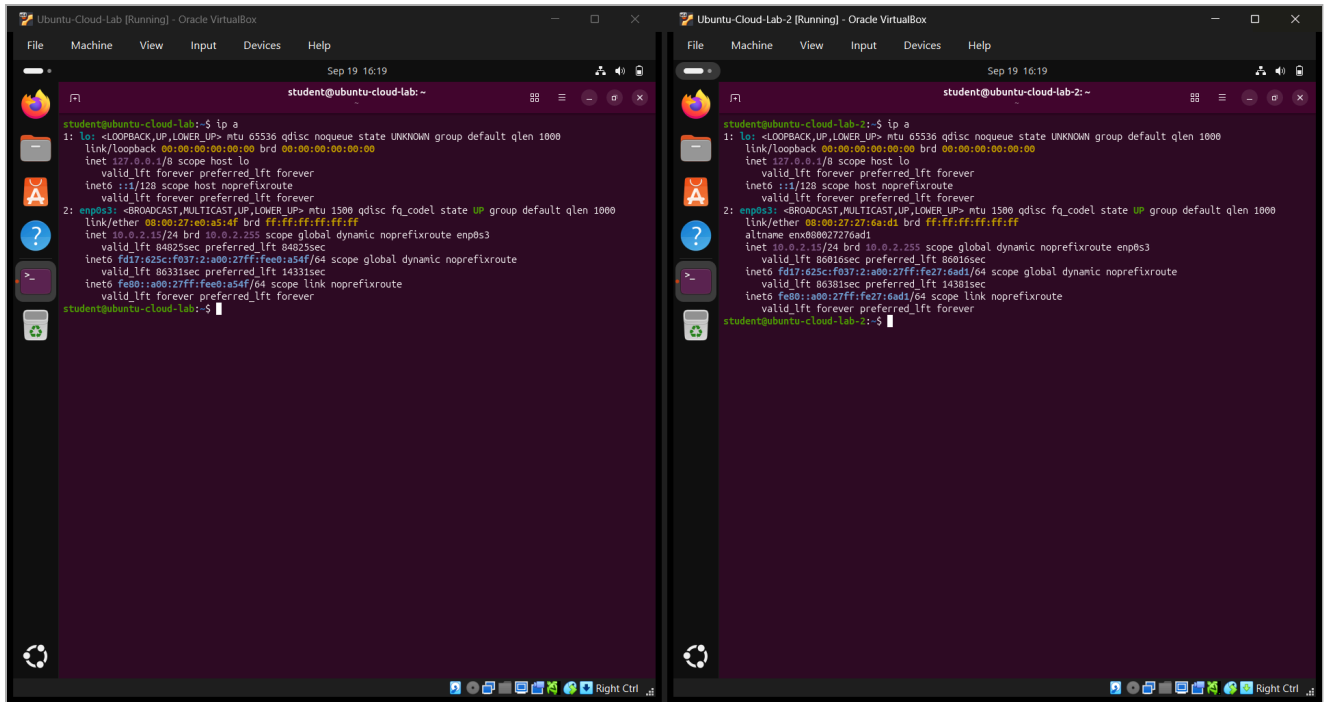


Figure 1: Both VMs on default NAT — `ip a` shows `10.0.2.15` on each

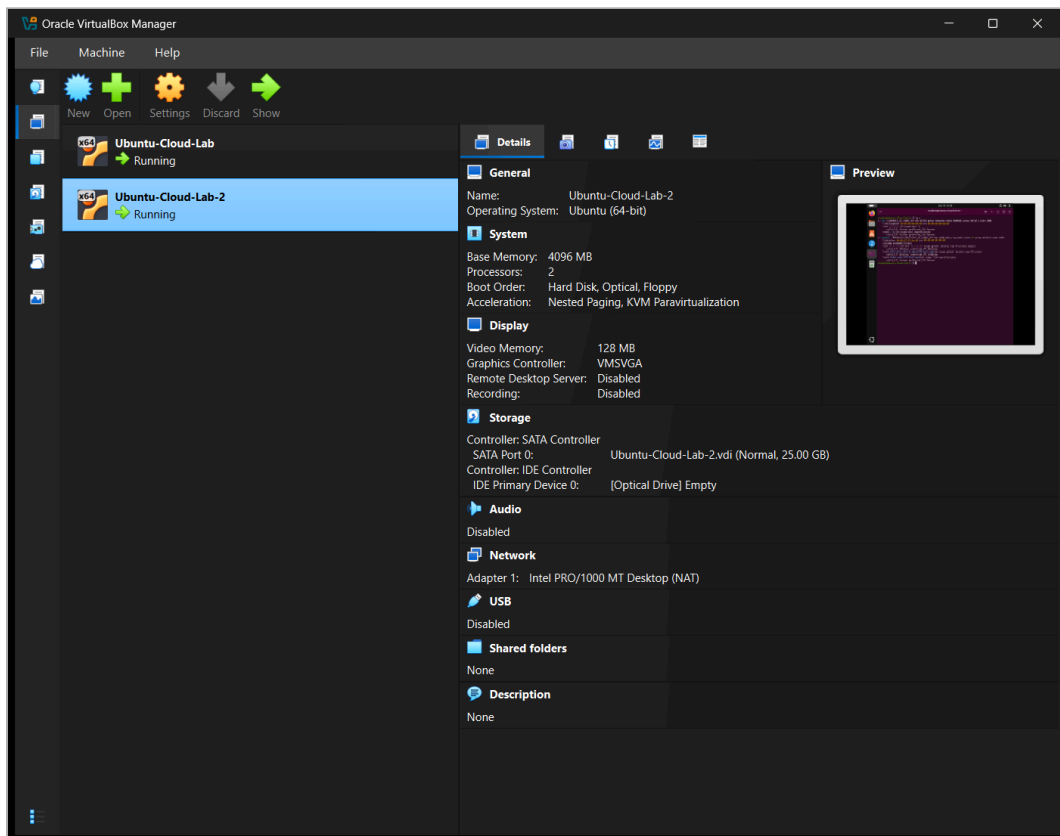


Figure 2: Two VMs in VirtualBox Manager

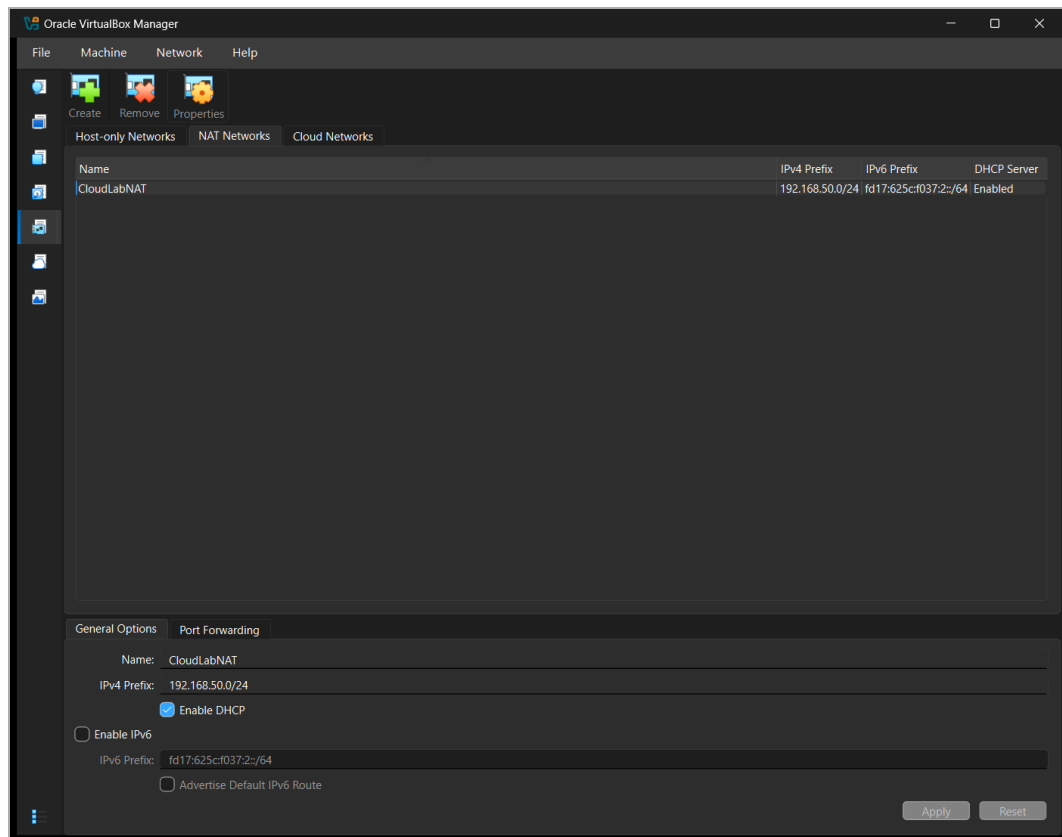


Figure 3: NAT Network *CloudLabNAT* (192.168.50.0/24, DHCP enabled) created in File → Tools → Network

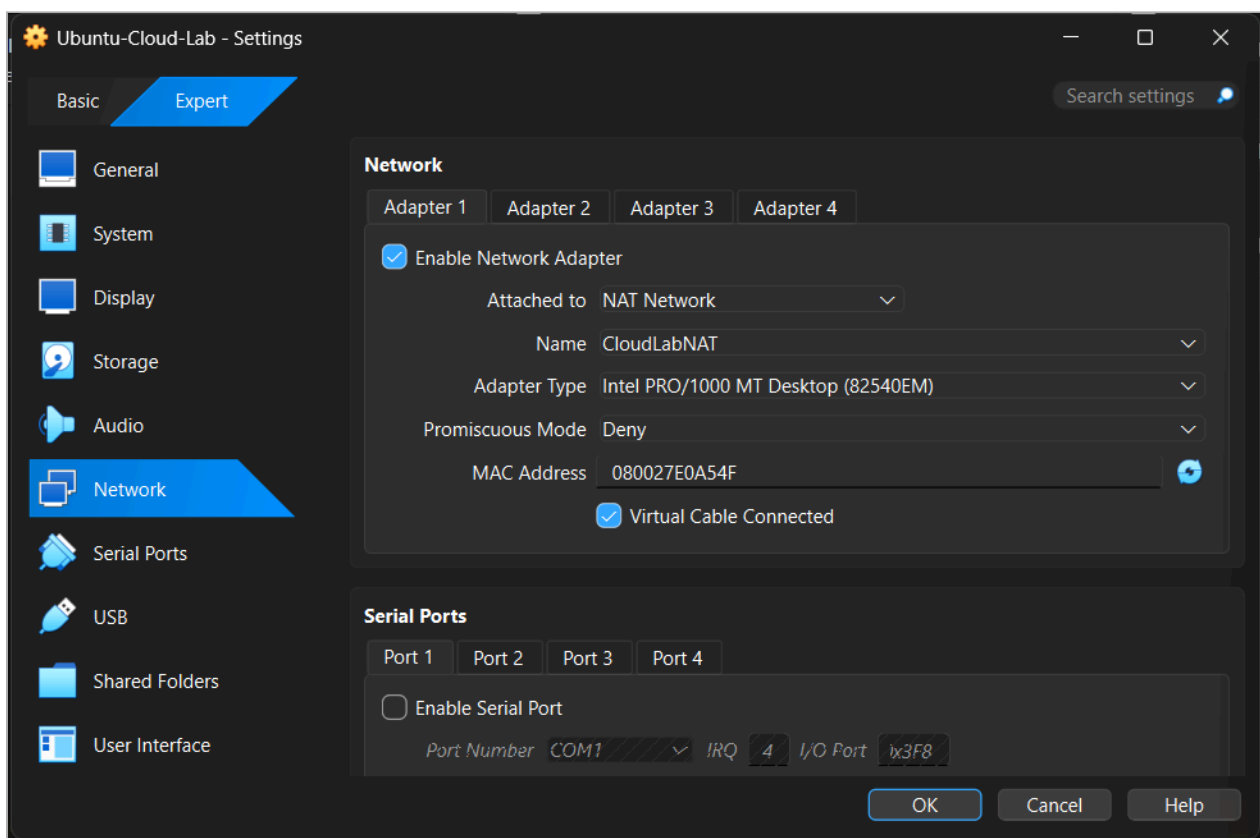


Figure 4: VM 1: Settings → Expert → Network → Attached to "NAT Network" → *CloudLabNAT*

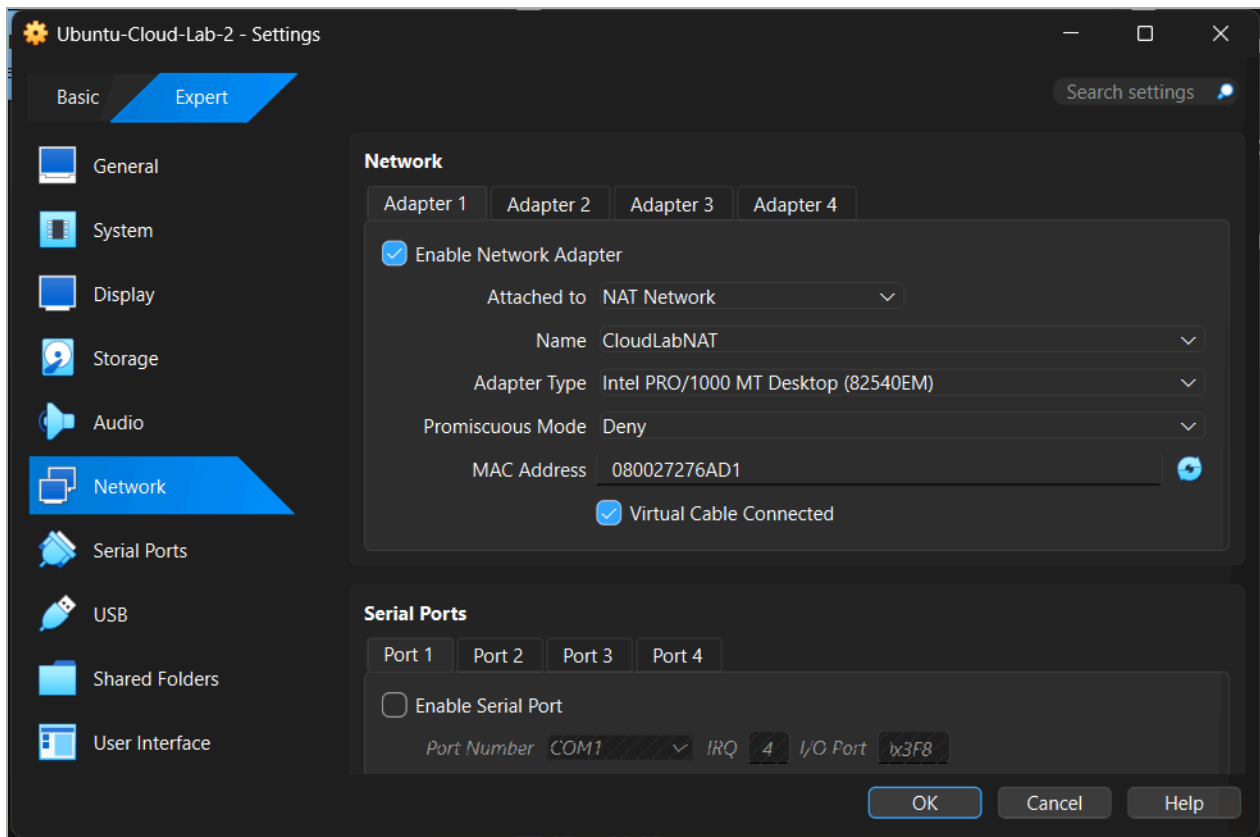


Figure 5: VM 2: Settings → Expert → Network → Attached to "NAT Network" → CloudLabNAT

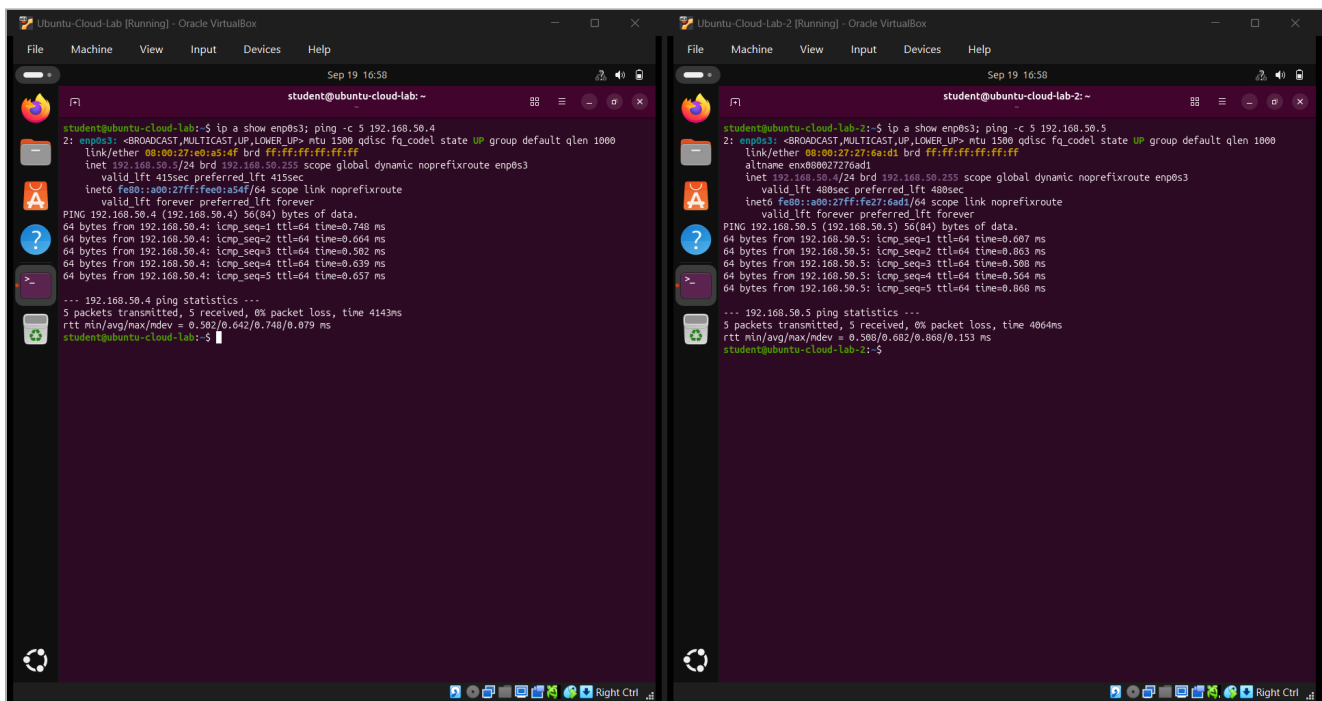


Figure 6: After restart: VM 1 (192.168.50.5) and VM 2 (192.168.50.4) ping each other with 0% packet loss

```
Sep 19 16:58
student@ubuntu-cloud-lab: ~
student@ubuntu-cloud-lab:~$ ip a show enp0s3; ping -c 5 192.168.50.4
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:e0:a5:4f brd ff:ff:ff:ff:ff:ff
    inet 192.168.50.5/24 brd 192.168.50.255 scope global dynamic noprefixroute enp0s3
        valid_lft 415sec preferred_lft 415sec
    inet6 fe80::a00:27ff:fe0:a54f/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
PING 192.168.50.4 (192.168.50.4) 56(84) bytes of data.
64 bytes from 192.168.50.4: icmp_seq=1 ttl=64 time=0.748 ms
64 bytes from 192.168.50.4: icmp_seq=2 ttl=64 time=0.664 ms
64 bytes from 192.168.50.4: icmp_seq=3 ttl=64 time=0.502 ms
64 bytes from 192.168.50.4: icmp_seq=4 ttl=64 time=0.639 ms
64 bytes from 192.168.50.4: icmp_seq=5 ttl=64 time=0.657 ms

--- 192.168.50.4 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4143ms
rtt min/avg/max/mdev = 0.502/0.642/0.748/0.079 ms
student@ubuntu-cloud-lab:~$
```

Figure 7: VM 1 → VM 2 ping (guest screen)

```
Sep 19 16:58
student@ubuntu-cloud-lab-2: ~
student@ubuntu-cloud-lab-2:~$ ip a show enp0s3; ping -c 5 192.168.50.5
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:27:6a:d1 brd ff:ff:ff:ff:ff:ff
    altname enx080027276ad1
    inet 192.168.50.4/24 brd 192.168.50.255 scope global dynamic noprefixroute enp0s3
        valid_lft 480sec preferred_lft 480sec
    inet6 fe80::a00:27ff:fe27:6ad1/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
PING 192.168.50.5 (192.168.50.5) 56(84) bytes of data.
64 bytes from 192.168.50.5: icmp_seq=1 ttl=64 time=0.607 ms
64 bytes from 192.168.50.5: icmp_seq=2 ttl=64 time=0.863 ms
64 bytes from 192.168.50.5: icmp_seq=3 ttl=64 time=0.508 ms
64 bytes from 192.168.50.5: icmp_seq=4 ttl=64 time=0.564 ms
64 bytes from 192.168.50.5: icmp_seq=5 ttl=64 time=0.868 ms

--- 192.168.50.5 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4064ms
rtt min/avg/max/mdev = 0.508/0.682/0.868/0.153 ms
student@ubuntu-cloud-lab-2:~$
```

Figure 8: VM 2 → VM 1 ping (guest screen)

8. Result

A custom NAT Network `CloudLabNAT` (`192.168.50.0/24`, DHCP enabled) was created in Oracle VirtualBox and both Ubuntu VMs were attached to it. After restarting, the VMs received unique addresses, `192.168.50.5` and `192.168.50.4`. Each VM successfully pinged the other with **5 packets transmitted, 5 received, 0% packet loss**.

9. Conclusion

With the default NAT mode every VM is isolated behind its own virtual router (all VMs get `10.0.2.15`) and cannot reach the others. Placing the VMs on a shared **NAT Network** puts them in one private subnet with a common gateway and DHCP server, so they can communicate with each other and still reach the Internet. This is the same model cloud providers use for instances inside a VPC subnet. The experiment also showed that cloned VMs need unique MAC addresses, and that reserved addresses (such as VirtualBox's DNS proxy at `.3`) must be kept out of the DHCP pool.